PROJECT REPORT ON

AI FORENSICS TIME MACHINE

An AI-Powered Cybersecurity Incident Response & Log Analysis Platform

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1. Executive Summary

The AI Forensics Time Machine is an advanced, production-grade cybersecurity platform designed to simplify the complex task of digital forensic log analysis. By leveraging modern AI models (Google Gemini) and real-time visualization techniques, the system enables security analysts to upload raw system logs, detect threats automatically, and reconstruct attack sequences through interactive timelines and graphs. This project bridges the gap between raw data and actionable intelligence, providing automated remediation recommendations and exportable forensic reports.

2. Introduction

In the modern digital landscape, large-scale systems generate millions of log entries daily. Manually auditing these logs to identify malicious behavior is nearly impossible for human analysts. Traditional SIEM tools often produce high noise and require significant manual configuration.

The AI Forensics Time Machine addresses these challenges by introducing an 'AI-First' approach to forensics. It utilizes a modular architecture to parse multi-format logs, identify suspicious patterns using rule-based heuristics, and uses LLMs to generate human-readable narratives of security incidents.

3. Project Objectives

- Automated Ingestion: Support various log formats (Linux, Windows, Network) with automated parsing.
- Threat Detection: Implement a rule engine capable of identifying 20+ types of cyberattacks.
- Visualization: Provide interactive Cytoscape-based attack graphs and Chart.js timelines.
- Forensic Replay: Enable step-by-step chronological playback of security incidents.
- AI Analysis: Integrate Google Gemini AI for generating executive summaries and remediation steps.

4. Technology Stack

Backend: Python, FastAPI, SQLite, SQLAlchemy, NetworkX, Google Gemini AI.

Frontend: React 19, Vite, Chart.js, Cytoscape.js, Vanilla CSS.

Data Flow:

1. Ingestion: Users upload log files or select live sources.
2. Parsing: Normalizes logs into structured Event format.
3. Detection: Engine scans events for threat signatures.
4. Reconstruction: Analyzer groups related events into Attack Chains.
5. AI Enrichment: AI Service processes data to generate summaries.

5. Methodology

The system employs a hybrid detection model combining rule-based signatures with AI-driven behavioral analysis. Rules cover Brute Force, Privilege Escalation, and Reconnaissance patterns. AI provides detailed narratives and prioritized remediation steps.

6. User Manual

1. Setup & Installation:

- Clone the repository and navigate to the project directory.
- Run 'pip install -r requirements.txt' in the backend directory.
- Run 'npm install' in the frontend directory.

2. Starting the Application:

- Start the backend server by running 'python -m uvicorn main:app --reload' in the backend directory.
- Start the frontend server by running 'npm run dev' in the frontend directory.

3. Using the Dashboard:

- Navigate to http://localhost:5173/ in your web browser.
- Upload your log files using the 'Upload Logs' button or choose a built-in pre-loaded dataset.
- View the interactive timeline, explore the attack graph, and replay security incidents.
- Navigate to the 'Analysis' section for an AI-generated forensic report and remediation recommendations.
- Export the findings as a PDF report.

7. Conclusion

In conclusion, the AI Forensics Time Machine demonstrates the power of combining digital forensics with generative AI, significantly reducing MTTD (Mean Time to Detect) and MTTR (Mean Time to Respond).